In [1]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

In [2]:
```
df = pd.read_csv("C:/Users/Kshitij/OneDrive/Desktop/DSBDAL/Assignment 3/Data/iris.c
df.head()
```

Out[2]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

In [3]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    str
dtypes: float64(4), str(1)
memory usage: 6.0 KB
```

In [4]:
```
print("Mean:\n", df.mean(numeric_only=True))
```

```
Mean:
 sepal_length    5.843333
sepal_width     3.054000
petal_length    3.758667
petal_width     1.198667
dtype: float64
```

In [5]:
```
print("Median:\n", df.median(numeric_only=True))
```

```
Median:
 sepal_length    5.80
sepal_width     3.00
petal_length    4.35
petal_width     1.30
dtype: float64
```

In [6]:
```
print("Mode:\n", df.mode(numeric_only=True))
```

```
Mode:
    sepal_length  sepal_width  petal_length  petal_width
0            5.0          3.0           1.5          0.2
```

In [7]:
```
print("Standard Deviation:\n", df.std(numeric_only=True))
```

```
Standard Deviation:
 sepal_length    0.828066
sepal_width     0.433594
petal_length    1.764420
petal_width     0.763161
dtype: float64
```

In [8]:
```
print("Minimum:\n", df.min(numeric_only=True))
print("Maximum:\n", df.max(numeric_only=True))
```

```
Minimum:
 sepal_length    4.3
sepal_width     2.0
petal_length    1.0
petal_width     0.1
dtype: float64
Maximum:
 sepal_length    7.9
sepal_width     4.4
petal_length    6.9
petal_width     2.5
dtype: float64
```

In [9]:
```
print(df.columns)
```

```
Index(['sepal_length', 'sepal_width', 'petal_length', 'petal_width',
       'species'],
      dtype='str')
```

In [10]:
```
for sp in df["species"].unique():
    print(f"\nStatistics for {sp}")
    print(df[df["species"]==sp].describe())
```

```
Statistics for setosa
       sepal_length  sepal_width  petal_length  petal_width
count      50.00000    50.000000     50.000000     50.00000
mean        5.00600     3.418000      1.464000      0.24400
std         0.35249     0.381024      0.173511      0.10721
min         4.30000     2.300000      1.000000      0.10000
25%         4.80000     3.125000      1.400000      0.20000
50%         5.00000     3.400000      1.500000      0.20000
75%         5.20000     3.675000      1.575000      0.30000
max         5.80000     4.400000      1.900000      0.60000

Statistics for versicolor
       sepal_length  sepal_width  petal_length  petal_width
count     50.000000    50.000000     50.000000    50.000000
mean       5.936000     2.770000      4.260000     1.326000
std        0.516171     0.313798      0.469911     0.197753
min        4.900000     2.000000      3.000000     1.000000
25%        5.600000     2.525000      4.000000     1.200000
50%        5.900000     2.800000      4.350000     1.300000
75%        6.300000     3.000000      4.600000     1.500000
max        7.000000     3.400000      5.100000     1.800000

Statistics for virginica
       sepal_length  sepal_width  petal_length  petal_width
count      50.00000    50.000000     50.000000     50.00000
mean        6.58800     2.974000      5.552000      2.02600
std         0.63588     0.322497      0.551895      0.27465
min         4.90000     2.200000      4.500000      1.40000
25%         6.22500     2.800000      5.100000      1.80000
50%         6.50000     3.000000      5.550000      2.00000
75%         6.90000     3.175000      5.875000      2.30000
max         7.90000     3.800000      6.900000      2.50000
```

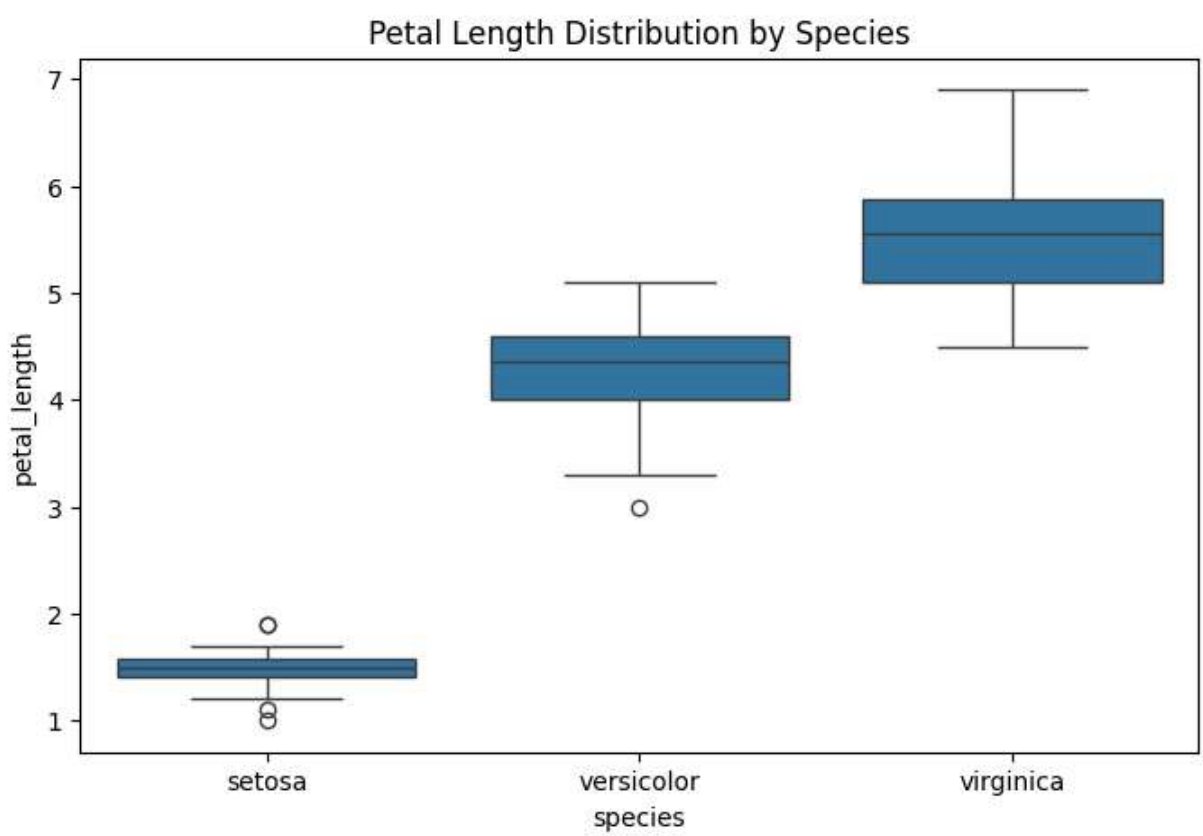
In [11]:

```python
grouped = df.groupby("species")
print(grouped.mean())
print(grouped.median())
print(grouped.std())
print(grouped.min())
print(grouped.max())
```

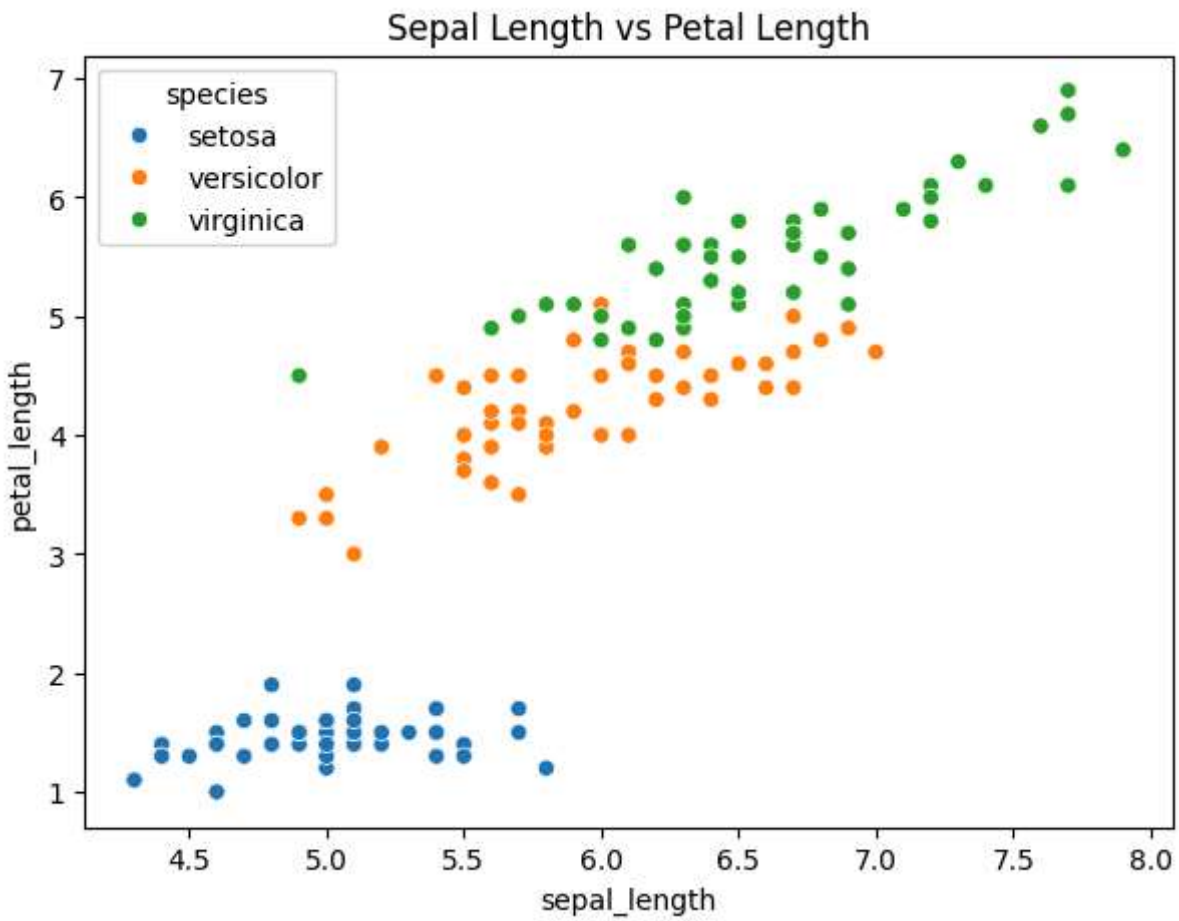
```
            sepal_length  sepal_width  petal_length  petal_width
species
setosa             5.006        3.418         1.464        0.244
versicolor         5.936        2.770         4.260        1.326
virginica          6.588        2.974         5.552        2.026
            sepal_length  sepal_width  petal_length  petal_width
species
setosa               5.0          3.4          1.50          0.2
versicolor           5.9          2.8          4.35          1.3
virginica            6.5          3.0          5.55          2.0
            sepal_length  sepal_width  petal_length  petal_width
species
setosa          0.352490     0.381024      0.173511     0.107210
versicolor      0.516171     0.313798      0.469911     0.197753
virginica       0.635880     0.322497      0.551895     0.274650
            sepal_length  sepal_width  petal_length  petal_width
species
setosa               4.3          2.3           1.0          0.1
versicolor           4.9          2.0           3.0          1.0
virginica            4.9          2.2           4.5          1.4
            sepal_length  sepal_width  petal_length  petal_width
species
setosa               5.8          4.4           1.9          0.6
versicolor           7.0          3.4           5.1          1.8
virginica            7.9          3.8           6.9          2.5
```

In [12]:

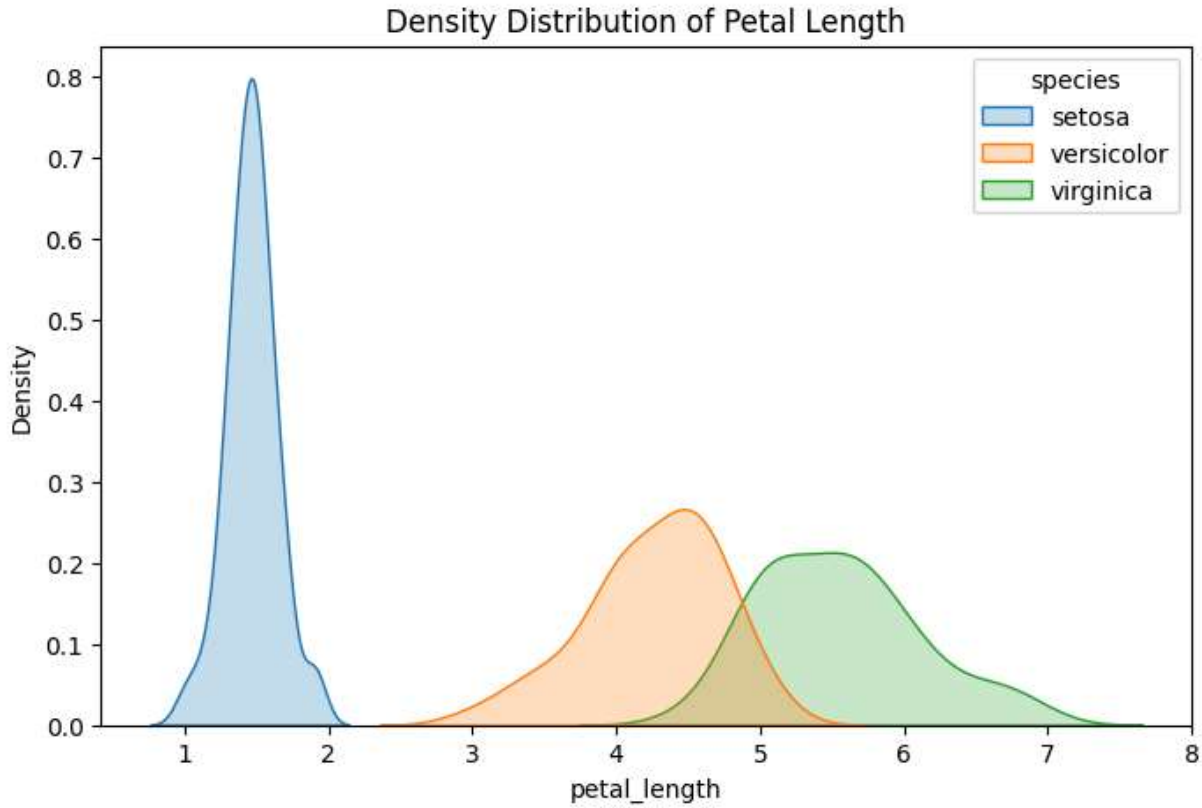
```
plt.figure(figsize=(8,5))
sns.boxplot(x="species", y="petal_length", data=df)
plt.title("Petal Length Distribution by Species")
plt.show()
```



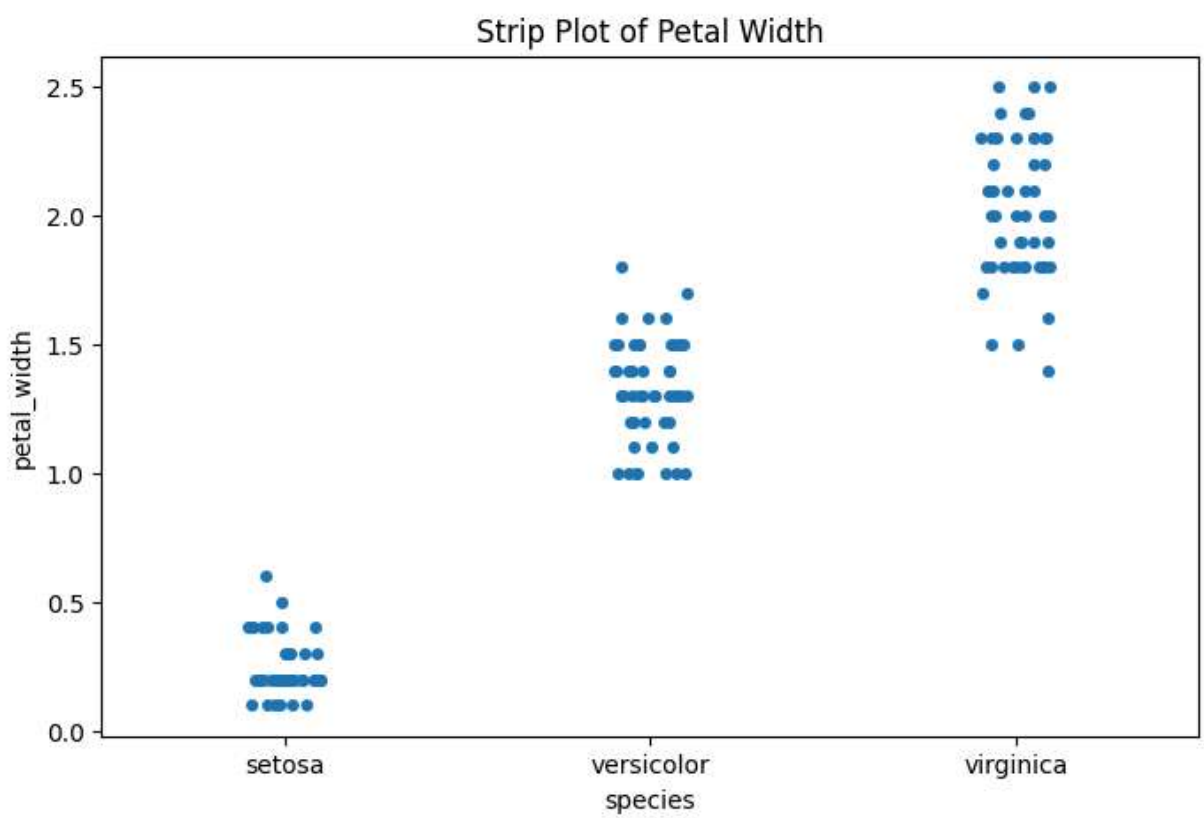
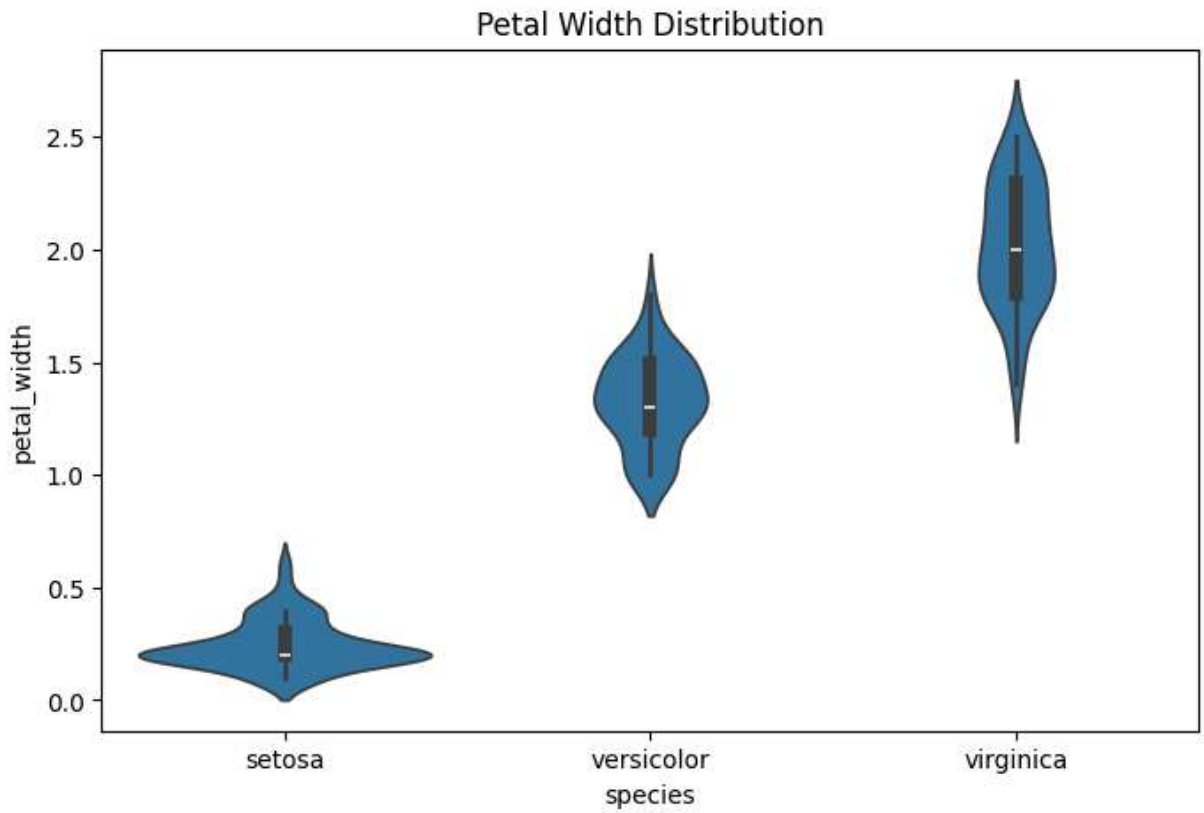
In [13]:
```
plt.figure(figsize=(7,5))
sns.scatterplot(
    x="sepal_length",
    y="petal_length",
    hue="species",
    data=df
)
plt.title("Sepal Length vs Petal Length")
plt.show()
```



In [14]:
```
plt.figure(figsize=(8,5))
sns.kdeplot(data=df, x="petal_length", hue="species", fill=True)
plt.title("Density Distribution of Petal Length")
plt.show()
```



In [15]:
```
plt.figure(figsize=(8,5))
sns.stripplot(x="species", y="petal_width", data=df, jitter=True)
plt.title("Strip Plot of Petal Width")
plt.show()
```



In [16]:
```
plt.figure(figsize=(8,5))
sns.violinplot(x="species", y="petal_width", data=df)
plt.title("Petal Width Distribution")
plt.show()
```